

Jampala Naresh

Agentic AI Solution Architect / Generative AI & LLM Lead / Data Scientist / Senior Machine Learning & Deep Learning Engineer



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LOCATION [Hyderabad, India](#)

Strategic AI leader with deep expertise in Agentic AI architectures, Generative AI platforms, LLM ecosystems, ML/DL predictive systems & cloud-native automation—seeking Deep Learning Engineer, Machine Learning Engineer, Data Scientist, Generative/Agentic AI Lead roles. Focused on driving enterprise-scale AI transformation by architecting intelligent, autonomous systems that enhance efficiency, reduce operational friction & accelerate innovation.

Domain Exposure: DRDO, IMD, CSIR, IT sector



PROFILE SUMMARY

- Strategic AI & Innovation Leader** with 9+ years steering enterprise transformation initiatives through Agentic AI, Generative AI, and predictive intelligence—delivering autonomous, business-aligned solutions across diverse industry verticals.
- Architect of High-Impact AI Systems**, driving end-to-end solution conceptualization, blueprinting, technical governance & multi-stakeholder orchestration to build scalable, reliable & measurable AI-driven business capabilities.
- Cross-Functional Leadership & Solution Ownership**, functioning at a Solution Architect / Senior Technical Lead level—leading requirement discovery, roadmap definition, cross-team alignment, and client-facing presales demos that accelerate adoption.
- Proven Enterprise Value Creation**, consistently delivering measurable outcomes such as drastic cycle-time reduction, enhanced decision intelligence, operational efficiency gains, and seamless automation across insurance, automotive, finance, defense, and weather domains.
- Business-Aligned Delivery Excellence**, known for translating complex business challenges into actionable AI solutions, ensuring high-impact outcomes, stakeholder satisfaction, and long-term organizational value through continuous improvement and innovation.
- Team Management expertise**, managing five-member cross-functional teams to ensure structured execution and quality outcomes.



CORE COMPETENCIES

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|-----------------------------|-------------------------------|--|
| ● Agentic AI Architecture | ● Generative AI Platforms | ● Large Language Models (LLMs) |
| ● Cloud-Native AI (AWS) | ● AI/ML Solution Architecture | ● Autonomous Multi-Agent Systems |
| ● AI Workflow Orchestration | ● Technical Leadership | ● Predictive Analytics & Deep Learning |
| ● Enterprise Automation | ● End-to-End AI System Design | ● MLOps & Model Deployment |



CERTIFICATIONS

- [AWS Certified Generative AI Developer - Professional \(AP1-C01\)](#) – Amazon Web Services
- [AWS Certified AI Practitioner \(AIF-C01\)](#) – Amazon Web Services
- [AWS Certified Machine Learning – Specialty \(MLS-C01\)](#) – Amazon Web Services
- [AWS Partner: Strands Agents Deep Dive – Completion Certificate | Advanced AI Agent Development using Strands SDK](#) – Amazon Web Services
- [Building Production-Ready AI Agents with Amazon Bedrock AgentCore – Completion Certificate](#) – Amazon Web Services
- [AWS Certified Generative AI Developer – Professional \(AIP-C01\)](#) – Amazon Web Services
- [Inside Agentic AI: Foundations and Frontiers](#) – Skillsoft
- [LLMs on Google Cloud: Embeddings, Function Calling, & Grounding with Gemini](#) – Skillsoft
- [Introduction to DeepSeek-R1 and Reasoning Models](#) – Skillsoft
- [An Introduction to Generative AI](#) – Skillsoft
- [Certified Secure Software Lifecycle Professional \(CSSLP\) 2019: Governance, Risk, and Compliance](#) – Skillsoft
- [Getting Started with Python: Introduction](#) – Skillsoft
- [Fundamentals of Deep Learning](#) – NVIDIA
- [Python Programming and Data Science](#) – National Institute of Electronics & Information Technology (NIELIT)
- Deep Learning & Onramp by MATLAB
- Technological Advancement in Healthcare: Role of Medical Image Processing with ML & DL



HONORS & AWARDS

- Senior Technical Lead driving Generative & Agentic AI innovations, including the “[Reimagining Claims Adjustment](#)” solution, contributing to Trianz being First Runner-Up in the [AWS Nova Challenge 2025](#) for excellence in AWS Nova LLM and Enterprise AI.
- Awarded the [Team Excellence Award \(Anytime Awards 2025\)](#) for high-impact GenAI and Agentic AI solutions
- Secured **State Rank 47** in **TSPGECET-2020** (M.Tech. entrance)



WHITE PAPER

- Developed an [AI-driven Intelligent Email Automation](#) system using Claude LLM to classify and route ~2M emails with ~95% accuracy, improving response time and operational efficiency by 60%.



IT SKILLS

Agentic AI, LLMs, Generative AI & Data Scientist

- Agentic AI Architecture & Orchestration:** CrewAI, AutoGen, AWS Strands-Agents, LangGraph, LangChain, Model Context Protocol (MCP), AWS AgentCore, Vertex AI Agent Builder
- Generative AI & LLM Ecosystem:** Claude, OpenAI, AWS Bedrock (FM Models, AgentCore, Data Automation), Ollama, RAG Pipelines, Prompt Engineering, LLMops, LLM Fine-Tuning, Retrieval-Augmented Systems
- Core Concepts & Certifications:** Fundamentals of Generative AI, Applications of Foundation Models, Responsible AI, Security & Governance for AI Solutions, Deep Reinforcement Learning

Machine Learning, Deep Learning & Algorithms

- ML/DL Algorithms:** Machine Learning Algorithms, IsolationForest, XGBoost, CNN, RNN, LSTM, ANN, DNN, Wavelets
- Frameworks:** TensorFlow, Keras, PyTorch, Transformer, HuggingFace, Scikit-learn, Python Programming & Data Science, MATLAB
- Data Science & Analytics:** Statistics, Data Analysis, Feature Engineering, Model Evaluation & Optimization

AWS Cloud Services: Amazon Bedrock, AgentCore, Bedrock Data Automation, SageMaker, Lambda, API Gateway, EC2, S3, DynamoDB, CloudWatch, Cognito, Step Functions, Glue, Athena, Redshift, Textract, EventBridge, SNS

Azure AI & Data Services: Azure AI Studio, Azure AI Foundry, Azure Databricks, Azure Functions, Azure Storage Blob, Event Grid, Azure Data Explorer

Google Cloud AI & Agentic Services: Vertex AI, Gemini Models, Vertex AI Agent Builder, Vertex AI Generative AI Studio, Vertex AI Model Garden, Vertex AI Embeddings, Vertex AI Vector Search, Vertex AI Extensions

Salesforce Integrations: GenAI & Agentic AI Integration, Bi-Directional Data Sync, Case Automation, CRUD Operations

Python, Frameworks & Web Development

- Python (3.13+): FastAPI, Django, Flask
- ML/DL Libraries: Scikit-learn, TensorFlow, Keras

SQL & Databases

- SQL Expertise: Advanced SQL, Query Optimization, Window Functions, Stored Procedures, Indexing Strategies
- Databases: MySQL, PostgreSQL, SQL Server

Snowflake: GenAI Tool Integration, Salesforce Sync, Read/Write Operations for Forecasting

Version Control: Git, Branching Strategies, Pull Requests, Code Reviews, Merge & Rebase

Other Technical Skills: Technical Project Leadership, Technical Documentation, Technical Support, Research, Microsoft Office, Excel



EDUCATION



M.Tech. – Systems & Signal Processing (SSP) | Pursuing
JNTUH College of Engineering, Hyderabad (Autonomous)
2022



B.Tech. – Electronics & Communication Engineering (ECE)
Jawaharlal Nehru Technological University (JNTU), Hyderabad
2013



CAREER TIMELINE



WORK EXPERIENCE

Trianz Digital Consulting Pvt. Ltd., Hyderabad | Feb'24 – Feb'26
Senior Technical Lead

- Spearheading an **Agentic AI-powered insurance claim adjustment system** using Prompt Engineering, AWS Strands-Agents, CrewAI, MCP, GPT-4o, AWS Bedrock AgentCore, and Vision AI agents, autonomously validating claim data, analyzing damage images, and assessing claim legitimacy with repair cost estimations, achieving 60% faster response times.
- Constructing a **fully automated Agentic AI pipeline** producing structured JSON outputs and professional PDF reports with AI-generated summaries, visual assessments, time logs, and embedded image evidence, streamlining the insurance claim lifecycle.
- Orchestrating **AI-driven email categorization and processing systems** using Amazon Bedrock (Claude & Llama LLM), Bedrock Data Automation, WorkMail, Lambda, and FastAPI, enhancing operational efficiency by 40%.
- Implementing **generative AI solutions for news summarization, social media content creation, and live script generation** leveraging FastAPI, AWS Bedrock (Claude & Llama LLM), S3, Transcribe, and Polly.

- Engineering a **real-time insurance assistance chatbot** for NYL using AWS Bedrock (Claude LLM), OpenSearch Serverless VectorDB, Lambda, and FastAPI, improving response accuracy by 90%.
- Developing an **AI-powered restaurant service chatbot** for Burger King franchise using AWS Bedrock (Claude & Llama LLM), S3, and OpenSearch VectorDB, streamlining order management and real-time service automation.
- Driving **AI-powered workflow automation** to reduce manual insurance processing effort from 10–15 days to 15 minutes, accelerating operational efficiency.
- Optimizing **AI-based decision-making & reporting processes** across insurance, restaurant & content automation projects, enabling faster insights, higher accuracy & scalable operations.

Technical Lead

- Architected a **GenAI-based email categorization system** using AWS Bedrock (Claude LLM), DynamoDB, and EventBridge, accelerating response times by 40% and optimizing email workflows.
- Engineered a **face recognition chatbot** leveraging AWS Rekognition, S3 & DynamoDB, strengthening identity verification & elevating user engagement.
- Devised an **AI-driven legislative bill summarization platform** using AWS Bedrock, Bedrock Data Automation, API Gateway, and Lambda, boosting document analysis speed by 50%.
- Developed a **sales forecasting model** integrated with Snowflake via AWS SageMaker, refining data pipelines & feature selection to enhance predictive accuracy.
- Automated **quality engineering processes** across the SDLC using AWS Lambda, AWS Bedrock (Claude LLM), OpenSearch, Generative AI, and FastAPI, enabling automated user stories, test case generation, and code analysis.
- Implemented a **scalable AI-powered email classification system with Salesforce integration**, leveraging AWS Bedrock (Claude & Llama LLM), Bedrock Data Automation, Prompt Engineering, Lambda, and FastAPI, continuously improving with user feedback.
- Enhanced **operational decision-making** by deploying AI solutions across multiple business functions, reducing manual interventions & improving efficiency.
- Optimized **predictive and automation workflows** across email processing, chatbot interactions, legislative document analysis, and sales forecasting, driving measurable gains in accuracy and throughput.

People Tech Enterprises Pvt Limited, Hyderabad, Telangana
Principal Data Scientist | May’22 – Feb’24

- Developed **AI/ML solutions for automotive applications** using Python frameworks such as Scikit-learn, XGBoost, and Hugging Face Transformers, incorporating IsolationForest, XGBoost, NLP techniques, and LLMs for predictive analytics and anomaly detection.
- Implemented **Generative AI models** on AWS SageMaker, Azure Databricks & other cloud platforms to enhance decision-making & operational efficiency.
- Optimized **vehicle monitoring & traffic sign detection**, improving detection accuracy by 94% & reducing claim processing time by 40%.
- Delivered **scalable AI solutions for Motors (GM) – US**, driving operational efficiency, cost savings, and improved service outcomes.
- Enhanced **predictive analytics & anomaly detection workflows**, enabling faster, accurate insights for automotive insurance & monitoring systems.

Teamware Solutions (QUANTUM LEAP CONSULTING PVT. LTD), Chennai
Senior Data Scientist | Feb’22 – May’22

- Developed **real-time event detection models** using IsolationForest for anomaly detection on large-scale datasets.
- Implemented **end-to-end ML pipelines** on Azure Databricks, Blob Storage, Stream Analytics, Functions & Event Grid for seamless data processing.
- Leveraged **Scikit-learn’s IsolationForest algorithm** to enhance anomaly detection accuracy and reduce false positives.
- Improved **system response time and predictive performance**, increasing operational efficiency by 90%.
- Optimized **cloud resource utilization**, achieving 35% cost savings and enhancing system scalability.

CSIR-National Geophysical Research Institute (NGRI), Hyderabad, Telangana
Project Associate II | Aug’19 – Jan’21

- Developed **predictive models and deep learning networks** using Python and MATLAB for seismic activity analysis.
- Applied **advanced data analysis and pattern recognition techniques** to enhance earthquake detection accuracy.
- Implemented **LSTM-based early warning systems** with TensorFlow/Keras for time-series prediction, improving predictive capabilities.
- Enhanced **early warning system efficiency by 90%**, reducing false alarms and strengthening overall earthquake preparedness.

IMD-Indian Institute of Tropical Meteorology (IITM), Pune, Maharashtra
Project Scientist B | Jan’18 – Apr’19

- Developed **deep learning networks** using MATLAB & Python for weather monitoring, enhancing predictive accuracy for atmospheric research.
- Performed **image classification of C & K Band Radar data** using the VGG19 model with TensorFlow/Keras, improving storm pattern recognition across three cloud image classes.
- Analyzed **PILS, ion chromatography (IC), and Rain Gauge data** to derive actionable insights for atmospheric studies.
- Optimized **data processing workflows**, achieving 30% faster analysis and improving weather prediction accuracy by 92%.

DRDO-Vehicles Research and Development Establishment (VRDE), Ahmednagar, Maharashtra
Research Fellow | May’15 – Jan’18

- Developed **high-performance, low-latency data processing systems** leveraging modern technologies to enable real-time analytics & efficient data throughput.
- Implemented **advanced deep learning techniques** to optimize system performance and enhance computational efficiency.
- Optimized **AI models for Automotive UAVs**, improving decision-making accuracy, navigation, and operational effectiveness.